Modern Academy
for Engineering and Technology in Maadi
Electrical Engineering

Academic Year	2025-2026
Semester	Fall
Exam Date	20 /1/2025
Academic Level:	Senior 2

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELCn425) Digital Signal Processing **Branch:** Communications & Computers
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan **Time:** 2 hours

Number of Pages: 1 Number of Questions: 4 Attempt all questions

Question 1 [10 Points] [LO's: 2, 4]

a- For LTIS define the following items: Stability, ROC and Difference Equation. 2

b- For the following transfer function:

$$H(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

(i) Plot the pole - zero diagram of H (z). 1
(ii) Find the difference equation of this filter, y (n). 2
(iii) Represent the filtering operation in signal flow diagram form. 1

c- Using residue method find "4" values for impulse response, h(n), of this system. 4

Question 2 [10 Points] [LO's: 1, 3]

a- Using Fig. s and Eq. s, define the following items: Correlation operation and End effect problem and its solution. 2

b- Given two sets of samples, $x_1 = \{1, 0, 1\}$ and $x_2 = \{0, 3, 1, 1\}$, calculate their cross-correlation, $r_{1,2}(j)$. 3

c- Find the output, y(n), of an electrical system of its impulse response function is, $h(n) = \{0, 0.8, 0.9, 1\}$ when the input sequence is, $x(n) = \{0, 2, 5, 1\}$ volts is applied. 5

Question 3 [10 Points] [LO's: 1, 2]

a- In detail, what are the main characteristics of the continuous time sinusoidal signals? 2.5

b- What are the digital filter design steps. 1.5

c- For the following signal, $x(t) = 10\cos(1000\pi t)$:

(i) If it is sampled with Fs = 2000 sample/sec, is it possible to reconstruct this signal back from its samples successfully? 1
(ii) Find the Nyquist frequency, F_N, for this signal. 1
(iii) Using the obtained Nyquist frequency, F_N find an expression for the corresponding sequence, x(n). 2
(iv) Find the values of x(n) in the first cycle. 2

Question 4 [10 Points] [LO's:2, 4]

a – Derive the sampling condition. 1

b- Compare between FIR and IIR filters. 1

c- For the following sequence: $x(n) = \{1, 0, 1, 1\}$:

(i) Using the DFT method, find the frequency components of this sequence. 3
(ii) Check your results obtained in part (i) by calculating x(n) using the IFFT algorithm. 3
(iv) Using your results, verify Parseval's theorem. 1
(v) What is the computational saving when using FFT algorithm instead of DFT algorithm? 1

Good Luck